

R documentation

of all in ‘./man’

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alpha_count	<i>Count Amino Acid Frequency in a FASTA File</i>
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Description

‘alpha_count’ counts the frequency of amino acids in a given FASTA file, optionally filtering sequences by length and simplifying the output.

Usage

```
alpha_count(file, seq_min = 0, seq_max, verbose = FALSE, simple = TRUE)
```

Arguments

file	A character string representing the path to the FASTA file.
seq_min	An integer specifying the minimum length of sequences to be counted. Default is 0.
seq_max	An integer specifying the maximum length of sequences to be counted. Default is 1e6.

verbose	Logical; if TRUE, prints the results to the console. Default is FALSE.
simple	Logical; if TRUE, removes amino acids with zero frequency from the output. Default is TRUE.

Details

This function reads a FASTA file and counts the occurrence of each amino acid. The results can be filtered by sequence length and simplified to exclude zero frequencies.

Value

If `simple = TRUE`, returns a named vector of non-zero frequencies. If `simple = FALSE`, returns a data frame with columns:

- Freq: Frequency of each amino acid.
- Prop: Proportion of each amino acid in the file.

Examples

```
# Example: Count amino acids in a FASTA file
# alpha_count("path/to/your/file.fasta")
```

area_shoelace	<i>Compute Area Using Shoelace Formula</i>
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Description

‘area_shoelace’ computes the area enclosed by a polygon defined by the points (x, y) using the shoelace formula (also known as Gauss’s area formula).

Usage

```
area_shoelace(x, y)
```

Arguments

x	A numeric vector representing the x-coordinates of the polygon vertices.
y	A numeric vector representing the y-coordinates of the polygon vertices.

Details

The shoelace formula calculates the area of a simple polygon whose vertices are described by their Cartesian coordinates in the plane. The vertices must be defined in order, either clockwise or counterclockwise.

Value

A numeric value representing the computed area.

Examples

```
# Example: Compute the area of a triangle
x <- c(2, 4, 6, 2)
y <- c(4, 4, 1, 4)
area_shoelace(x, y)
```

auto_test

Perform Automatic Statistical Test

Description

This function automatically selects and performs the appropriate statistical test based on the input data. It first determines whether to use a chi-squared test for a 2x2 matrix. For non-matrix inputs, it checks for normality and chooses between a t-test (if both samples are normally distributed) or a Wilcoxon signed-rank test (if one or both samples are not normally distributed).

Usage

```
auto_test(x, y, paired = FALSE)
```

Arguments

x	A numeric vector or a 2x2 matrix. If x is a matrix, a chi-squared test is performed.
y	A numeric vector. This parameter is required unless x is a matrix.
paired	Logical; if TRUE, indicates that the data are paired.

Details

The function performs the following checks:

- If x is a 2x2 matrix, a chi-squared test is performed.
- If all values in x or y are the same, a message is displayed and 0 is returned.
- For numeric vectors, normality is tested using the Shapiro-Wilk test. If both vectors are normally distributed, an independent or paired t-test is performed based on the paired argument.
- If normality is violated, the Wilcoxon signed-rank test is used.

Value

The result of the statistical test as an object of class "htest". If the values in x or y are all the same, the function returns 0 and prints a message.

Examples

```
# Example usage with a 2x2 matrix for chi-squared test
matrix_data <- matrix(c(10, 20, 30, 40), nrow = 2)
auto_test(matrix_data)

# Example usage with numeric vectors
x <- rnorm(10)
y <- rnorm(10)
auto_test(x, y)
```

delete_files	<i>Delete files and folders with a specified interval</i>
--------------	---

Description

This function deletes files and folders from a specified directory with a specified number and interval.

Usage

```
delete_files(folder_path, num_to_delete, interval_sec, show_more)
```

Arguments

folder_path	The path to the folder to delete contents from.
num_to_delete	The number of files/folders to delete at each interval.
interval_sec	The interval in seconds between each delete operation.
show_more	show more information of the deleted files.

Value

None.

eGFR	<i>Calculate eGFR (Estimated Glomerular Filtration Rate)</i>
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Description

Calculates the estimated Glomerular Filtration Rate (eGFR) based on serum creatinine levels, age, sex, and units (mg/dL or umol/L). The formula is modified specifically for Chinese patients based on the research by Ma et al. (2006).

Usage

```
eGFR(pcr, age, sex, Units = c("mg/dL", "umol/L"))
```

Arguments

pcr	Numeric vector representing serum creatinine levels. The unit is specified by Units.
age	Numeric vector representing the age of the individual(s) in years.
sex	Character vector representing the sex of the individual(s). Accepted values are "Male" or "Female".
Units	Character; specifies the unit of serum creatinine levels. Accepted values are "mg/dL" or "umol/L".

Details

The eGFR is an important measure of kidney function. This function calculates eGFR using a formula that is specifically modified for Chinese patients:

$$eGFR(ml/min \text{ per } 1.73m^2) = 186 \times Pcr^{-1.154} \times Age^{-0.203} \times 1.227 \times 0.742(if \text{ female})$$

The function accepts serum creatinine levels (Pcr), age, sex, and the units of creatinine measurement (mg/dL or umol/L). The formula is adjusted for sex, multiplying by 0.742 for females.

Value

A numeric vector of eGFR values (ml/min per 1.73 m²).

Note

If the units are specified as "umol/L", the function converts these to "mg/dL" before applying the formula. The conversion is done using the factor 1 mg/dL = 88.4 umol/L.

References

Ma, Y. et al. (2006). Modified glomerular filtration rate estimating equation for Chinese patients with chronic kidney disease. Journal of the American Society of Nephrology, 17(10), 2937-2944.

Examples

```
# Example usage:
n <- 1e3
pcr <- rnorm(n, 50, 3)
age <- runif(n, 40, 70)
sex <- sample(c("Female", "Male"), n, TRUE)
egfr <- eGFR(pcr, age, sex, Units = "mg/dL")
egfr2 <- 186 * (pcr^(-1.154)) * (age^(-0.203)) * 1.227 * ifelse(sex == "Female", 0.742, 1)
all.equal(egfr, egfr2)
```

error_to_character	<i>error_to_character</i>
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Description

Convert error object to character string.

Usage

```
error_to_character(e)
```

Arguments

e	Error object.
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file_type_analysis	<i>Analyze File Types in a Directory</i>
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Description

‘file_type_analysis’ analyzes the types of files within a specified directory and generates a bar plot showing the percentage distribution of each file type.

Usage

```
file_type_analysis(directory)
```

Arguments

directory	Character; the path to the directory to analyze.
-----------	--

Value

A list (invisibly) containing:

- files_type_freq: A data frame summarizing file types and their frequency.
- files_name: A list of file names grouped by their types.

Examples

```
# Example: Analyze file types in a specified directory  
  
file_type_analysis(R.home())
```

identify_mode	<i>calculate mode</i>
---------------	-----------------------

Description

calculate mode

Usage

```
identify_mode(x)
```

Arguments

x	numeric vector
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matrix_test	<i>Perform Group-wise Statistical Tests</i>
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Description

The ‘matrix_test’ function performs statistical tests between groups in a data frame. It can conduct either chi-squared tests (if ‘chisq_test = TRUE’) or other statistical tests provided by the user through the ‘test’ parameter (default is ‘auto_test’).

Usage

```
matrix_test(dat, group = "Group", chisq_test = FALSE, test = auto_test)
```

Arguments

dat	A data frame containing the data to be tested.
group	A character string specifying the name of the column in ‘dat’ that contains the group labels. Default is "Group".
chisq_test	Logical; if TRUE, the function performs a chi-squared test between groups. Default is FALSE.
test	A function specifying the statistical test to use for non-chi-squared tests. Default is auto_test.

Value

A data table (if using ‘data.table::rbindlist’) summarizing the test results, including:

- x: The first group in the comparison.
- y: The second group in the comparison.
- p: The p-value of the test.
- method: The method used for the test.
- df: Degrees of freedom (if applicable).
- statistic: The test statistic.

- mean_x: Mean of the first group.
- sd_x: Standard deviation of the first group.
- mean_y: Mean of the second group.
- sd_y: Standard deviation of the second group.

Examples

```
# Example 1
n <- 1e3
dat <- data.frame(
  Group = paste0("Group_", 1:100) |> rep(each = 10),
  GLD = rnorm(n, 3, 2),
  W0J = rpois(n, 3),
  W0K = rnorm(n, 2, 4)
)
suppressWarnings(
  ret1 <- matrix_test(dat, group = "Group")
)
ret1

# Example 2
dat <- data.frame(
  Type = paste0("Group_", 1:100) |> rep(each = 10),
  Smoking = rpois(n, 2),
  nonSmoking = rpois(n, 5)
) |>
  aggregate(. ~ Type, data = _, sum)

ret2 <- matrix_test(dat, group = "Type", chisq_test = TRUE)
ret2
```

prof_check

Profile a Function's Performance

Description

‘prof_check’ profiles the performance of a specified function, capturing both memory usage and execution time. It utilizes R’s built-in ‘Rprof’ tool for profiling.

Usage

```
prof_check(fun, ...)
```

Arguments

fun	A function to be profiled.
...	Additional arguments to pass to the function specified in ‘fun’.

Value

The result of the profiled function call (invisibly).

Examples

```
# Example: Profile the performance of the eGFR function
n <- 1e3
pcr <- rnorm(n, 50, 3)
age <- runif(n, 40, 70)
sex <- sample(c(0L, 1L), n, TRUE) |> factor(labels = c("Female", "Male"))
prof_check(eGFR, pcr, age, sex, "mg/dL")
```

showProgress

*Show Progress***Description**

‘showProgress’ is used to display the progress of a process in the console.

Usage

```
showProgress(percentage, value)
```

Arguments

percentage	Numeric; the percentage of completion (between 0 and 1).
value	Character; a string representing the current progress state.

Examples

```
# Example: Show progress at 82%
showProgress(0.82, "82%")
```

zero_or_value

*zero_or_value***Description**

Return zero if the expression encounters an error, or return the value.

Usage

```
zero_or_value(expre, zero = NA_real_, verbose = FALSE)
```

Arguments

expre	Expression to be evaluated.
zero	Return value if the expression encounters an error. If NULL, returns the error message as a character string.
verbose	Logical indicating whether to display error messages.

Examples

```
zero_or_value(1 + "a", verbose = TRUE) # Displays error message  
zero_or_value(1 + 2) # Returns 3  
zero_or_value(1 + "a") # Returns 0 without displaying error message  
zero_or_value(1 + "a", zero = NULL)
```

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